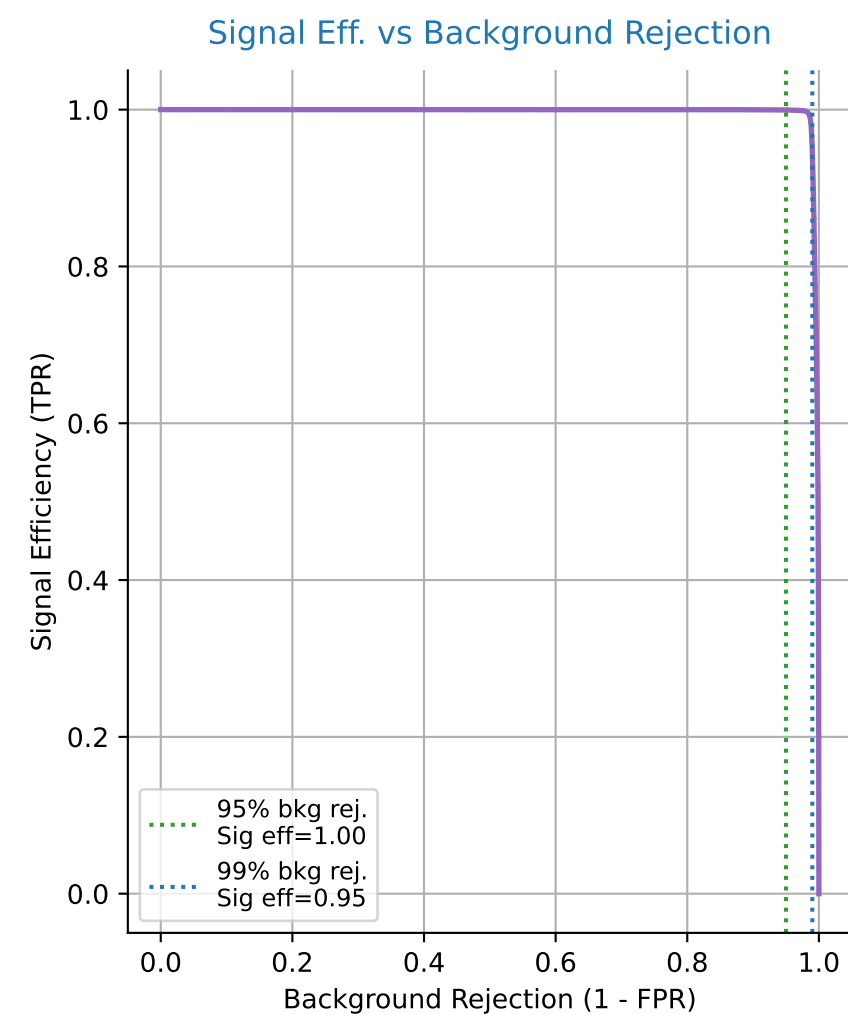
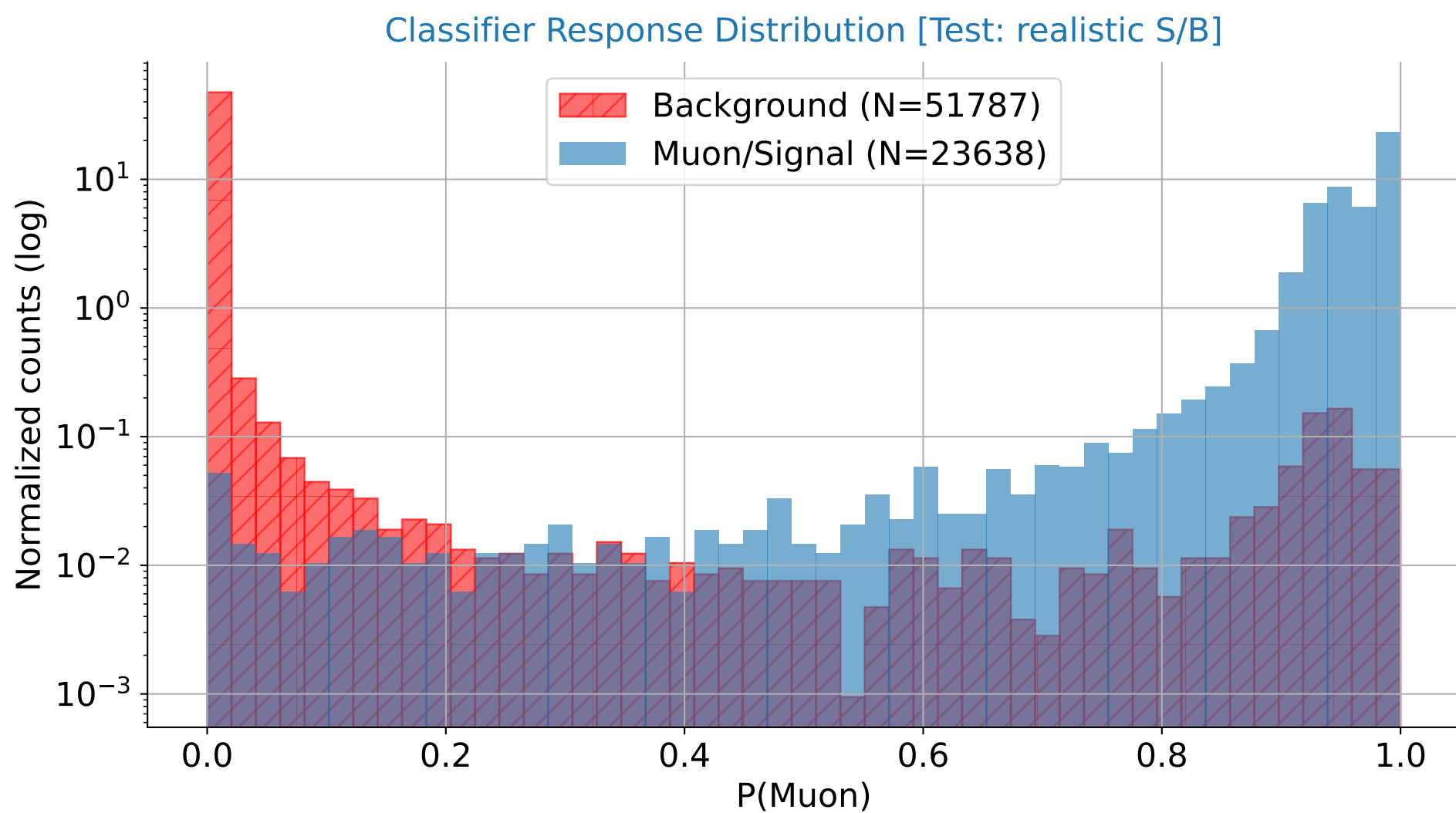
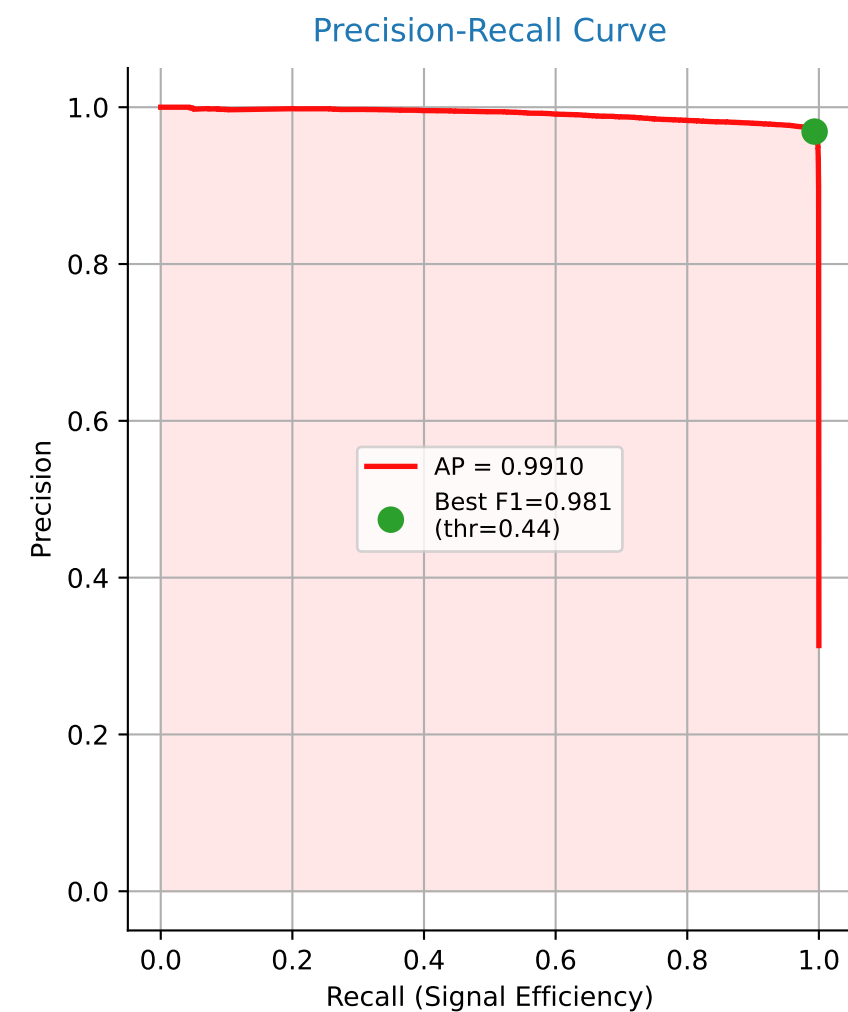
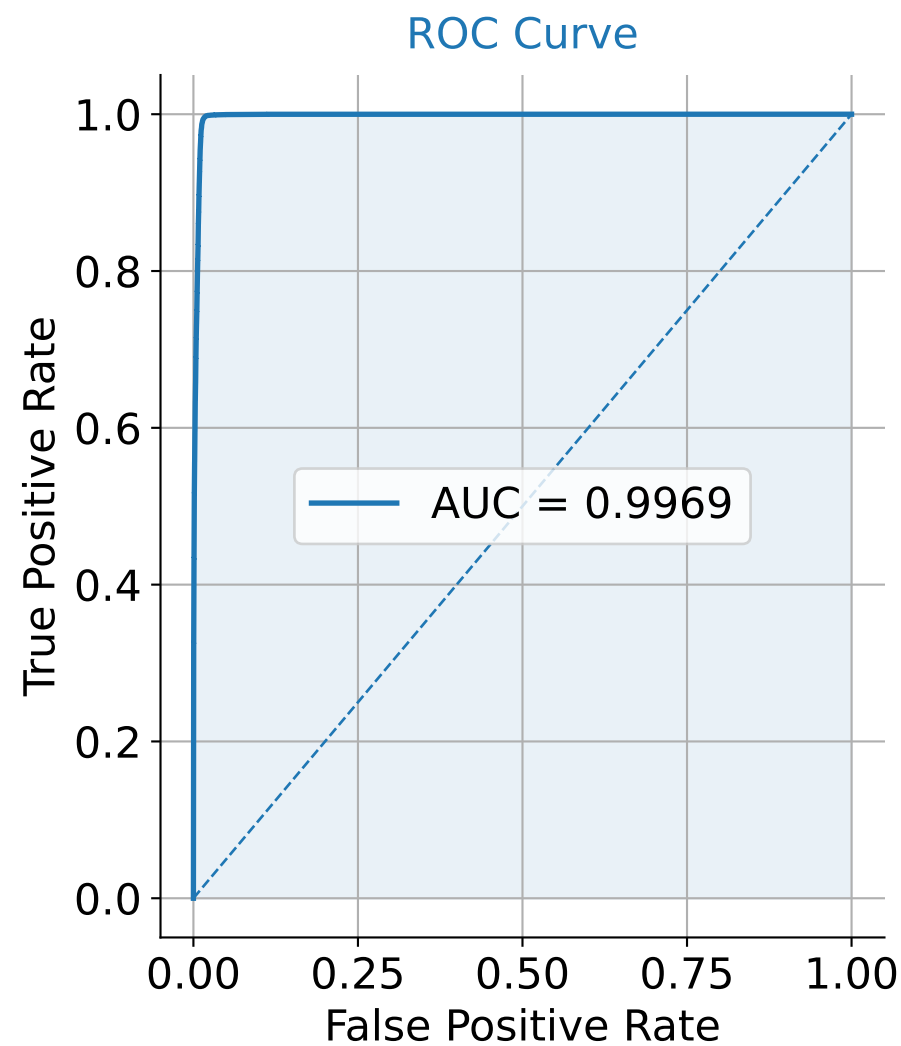
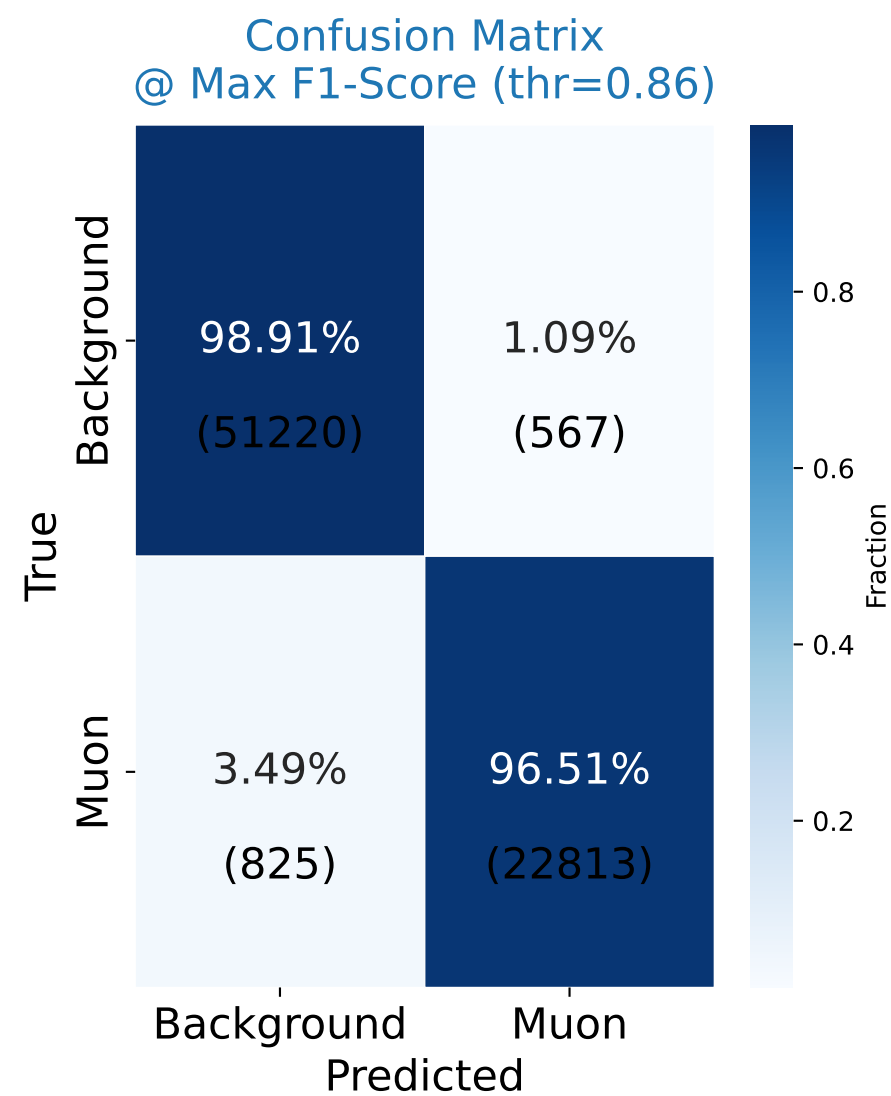
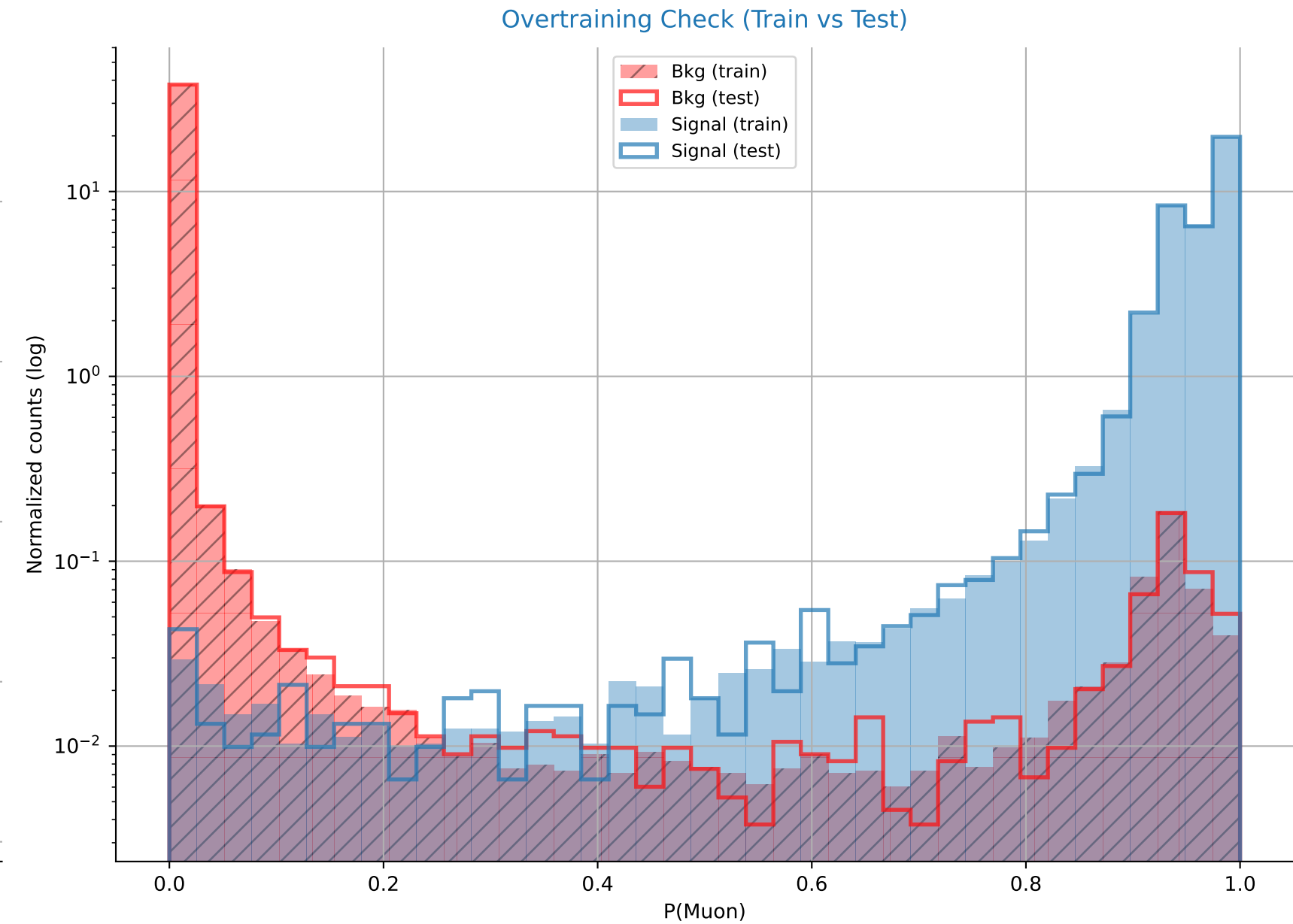
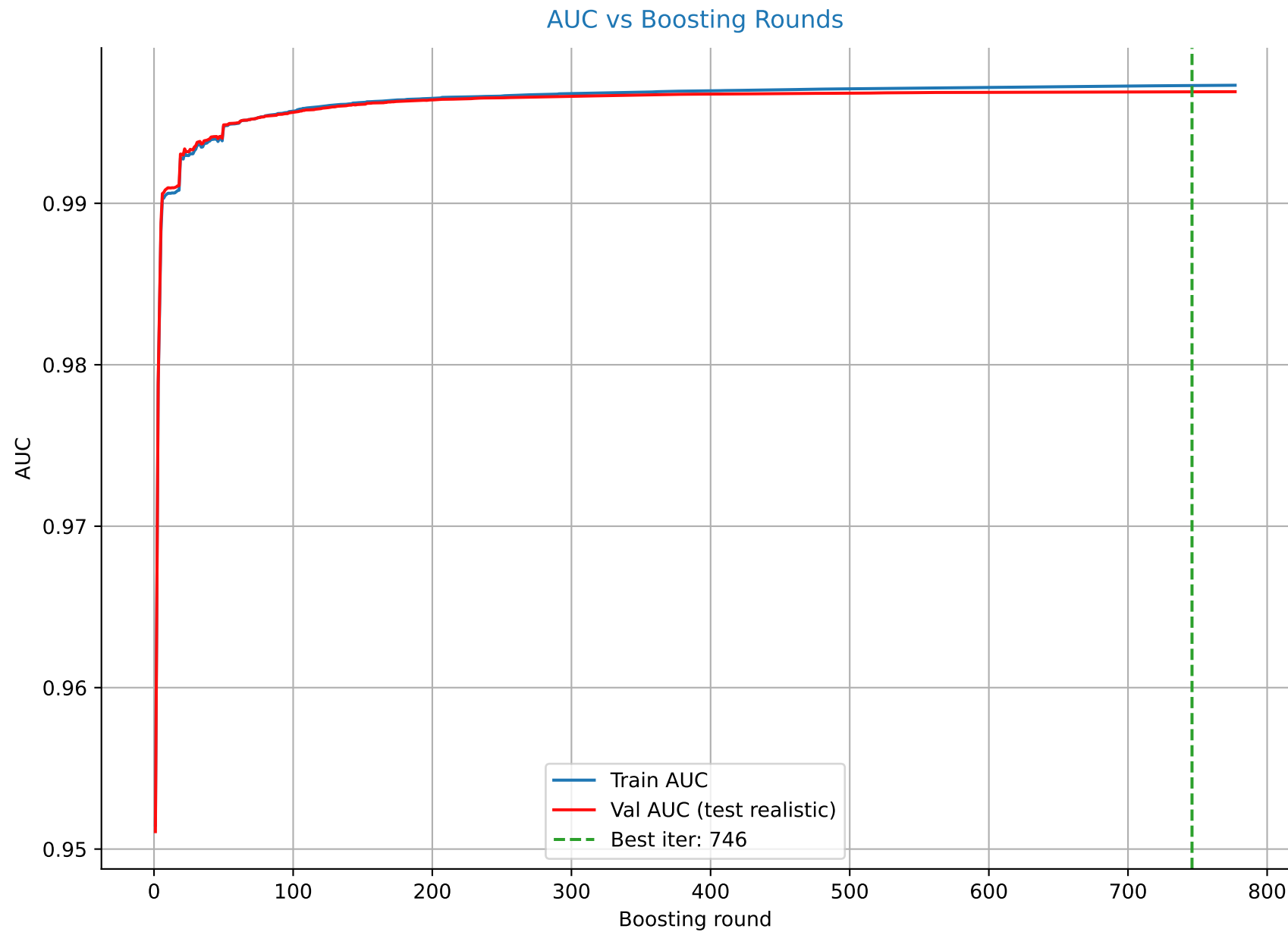


Muon Identification — XGBoost Analysis

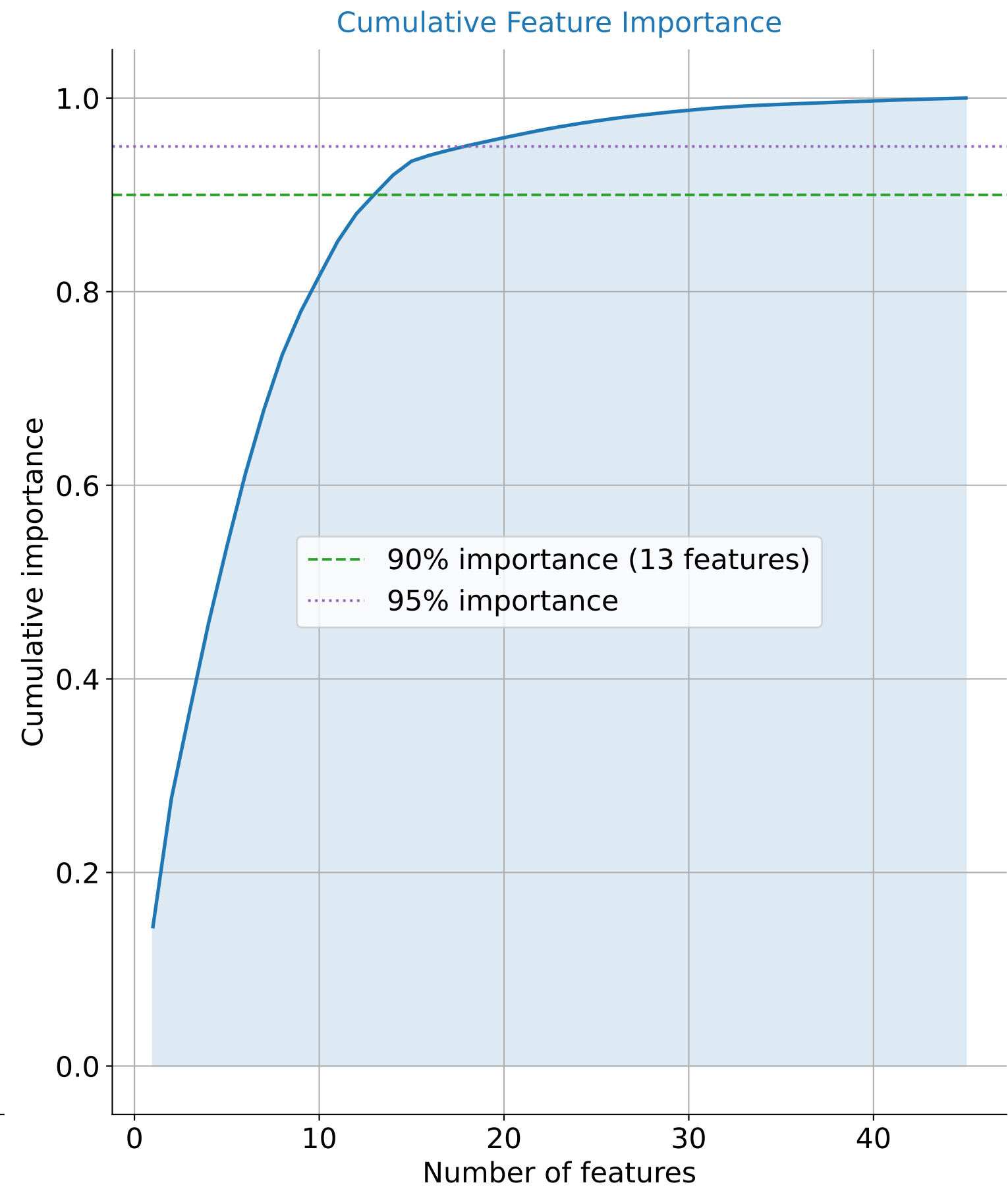
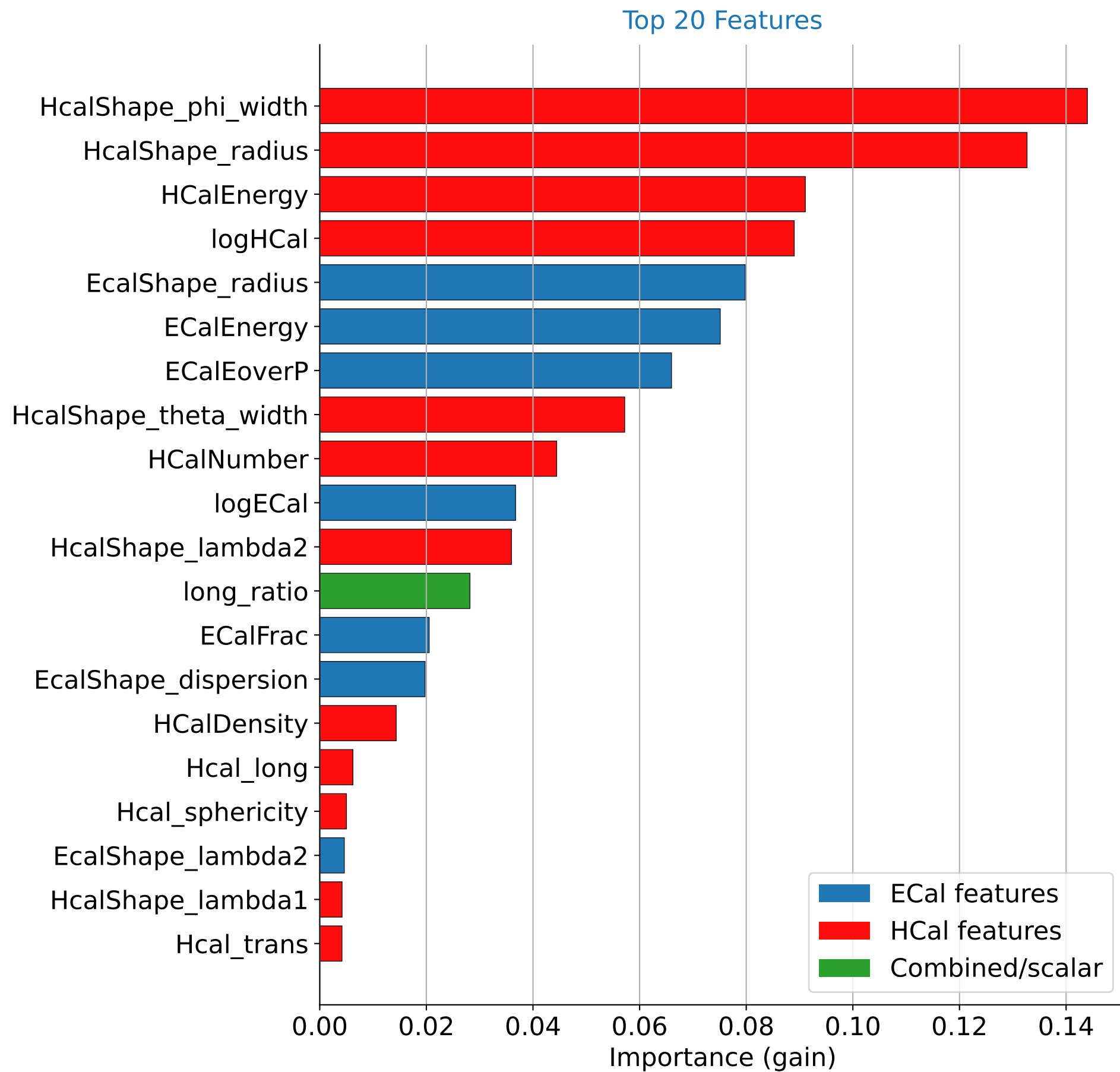
[Train S/B=1:2 | Test S/B=1:2 (realistic, all files)]



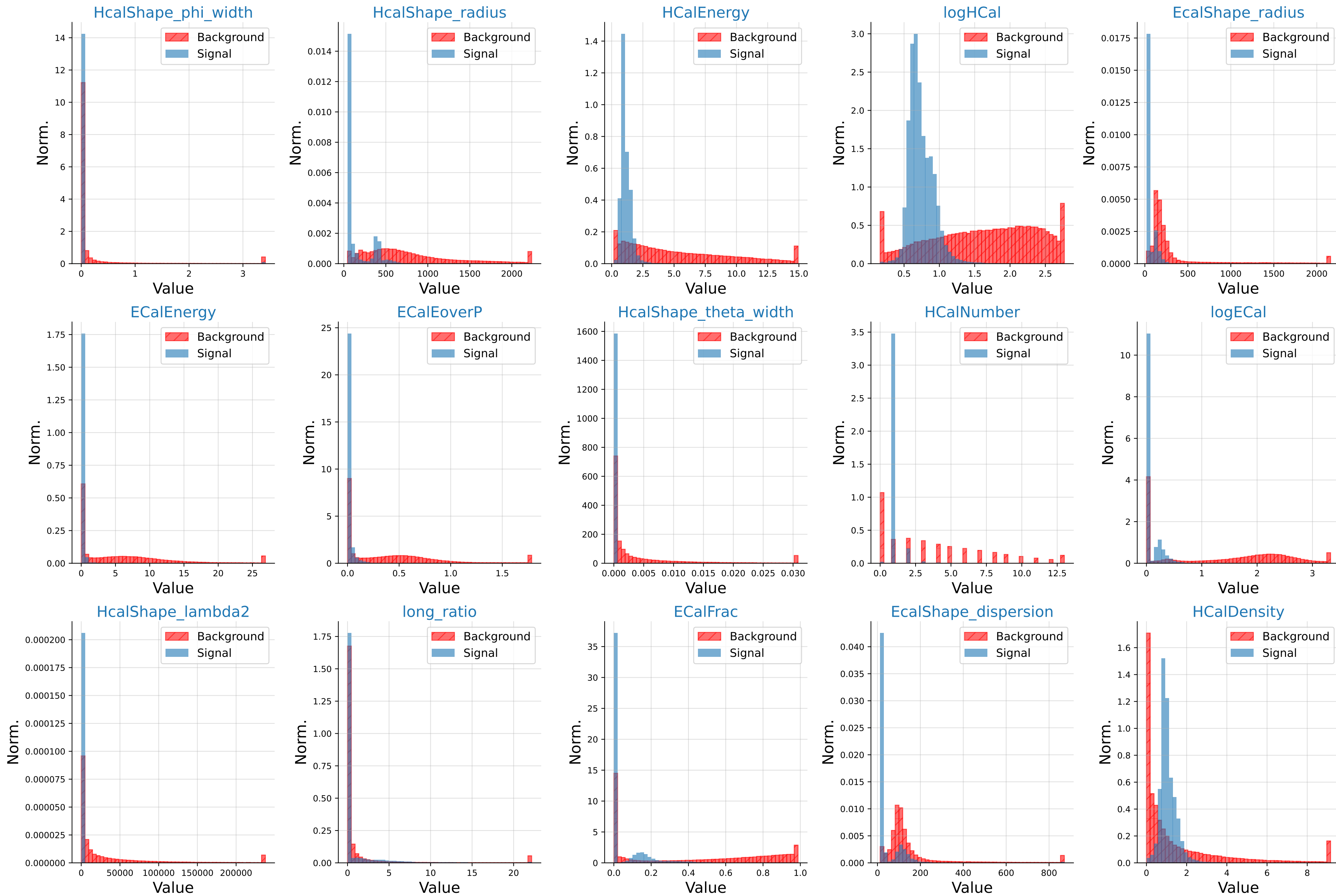
XGBoost Training Diagnostics



Feature Importances — XGBoost

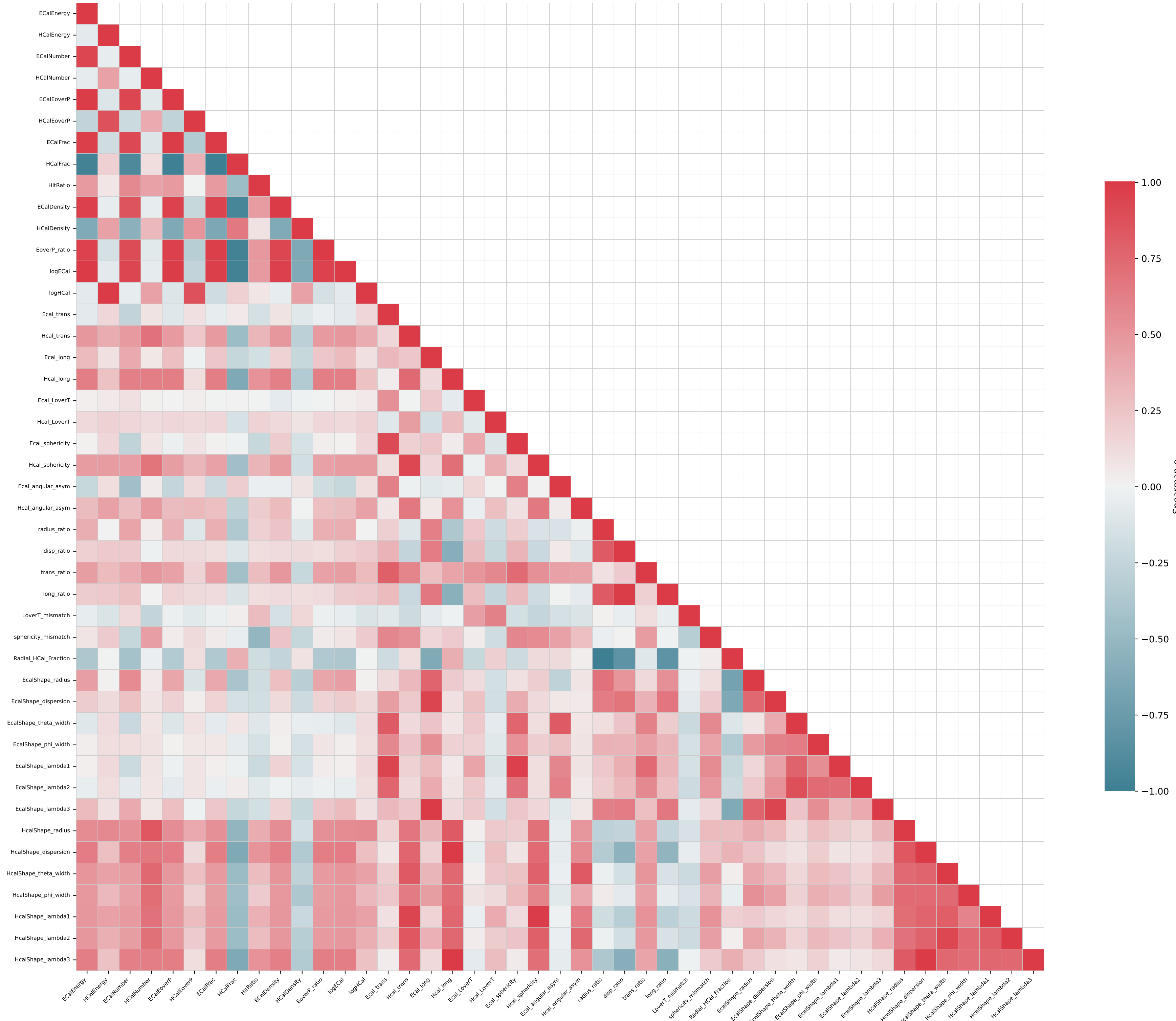


Key Physics Variable Distributions: Signal vs Background
[rozkłady z całego datasetu przed podziałem]

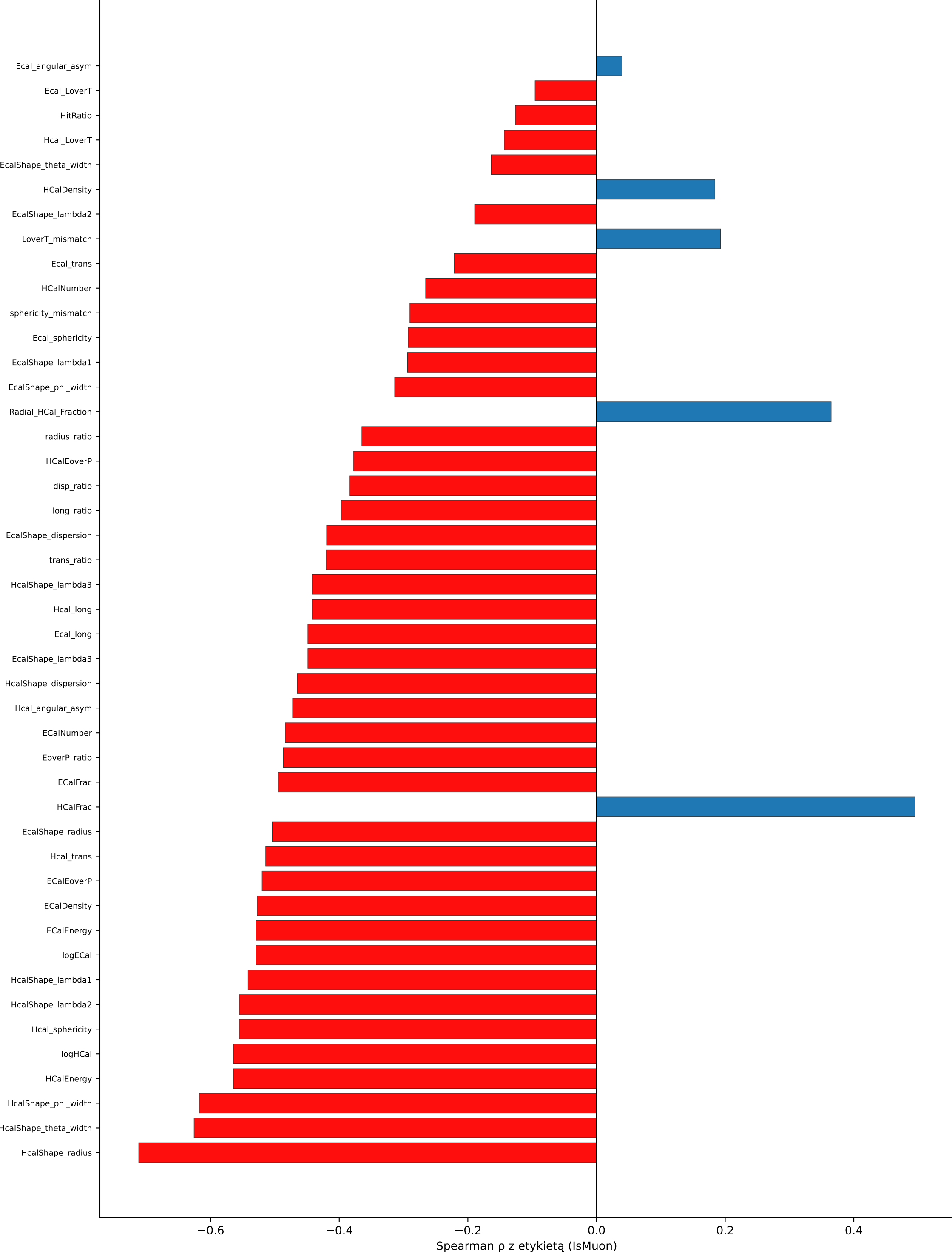


Spearman Correlation Matrix — All Features
(wartości z całego datasetu)

Wyższe $|\rho| \rightarrow$ silniejsza monotonna zależność między cechami



Korelacja cech z klasą sygnał/tło
Niebieski = pozytywna (→muon), Czerwony = negatywna (→tło)

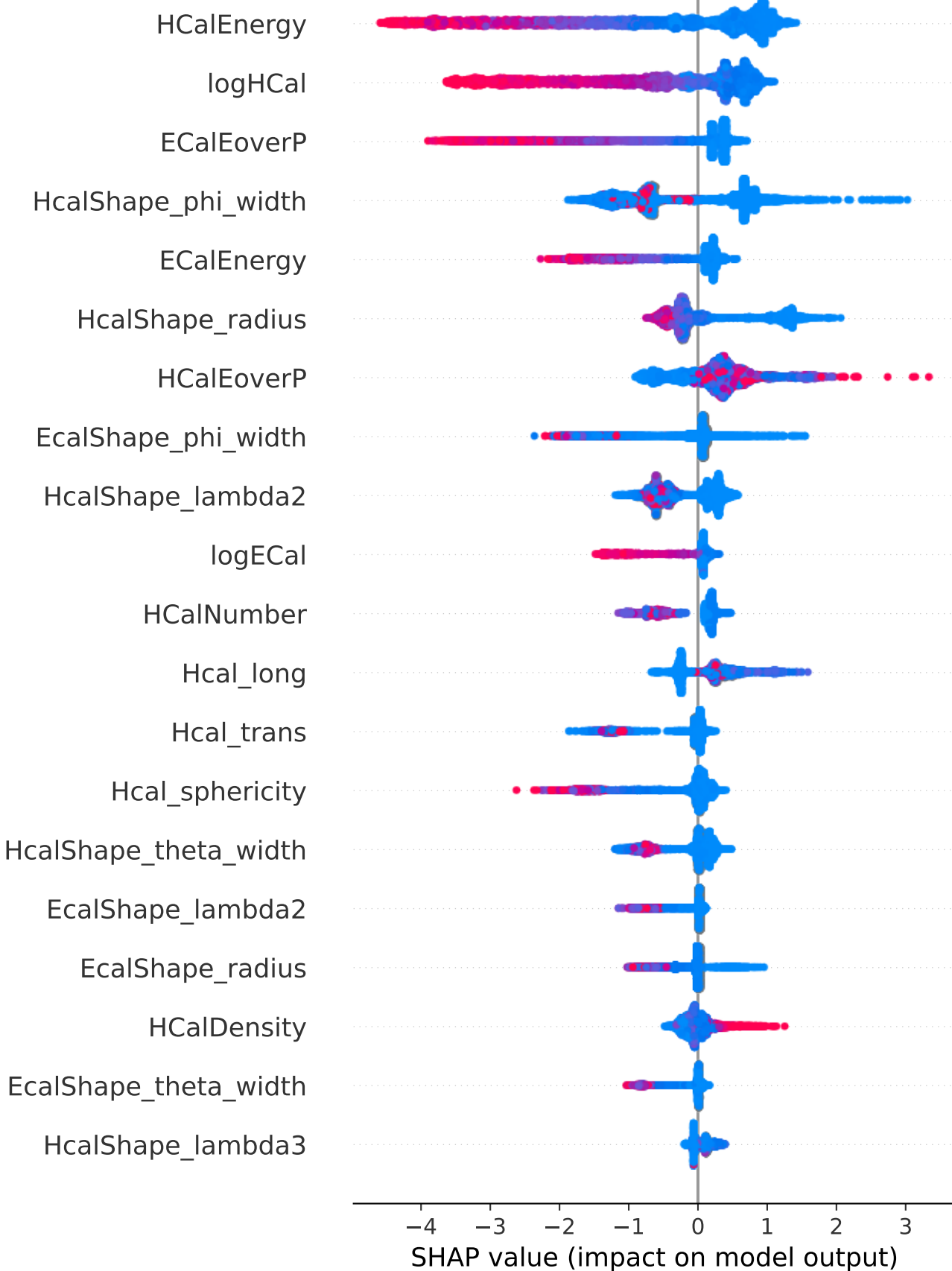


SHAP Summary Plot — Feature Impact on P(Muon)

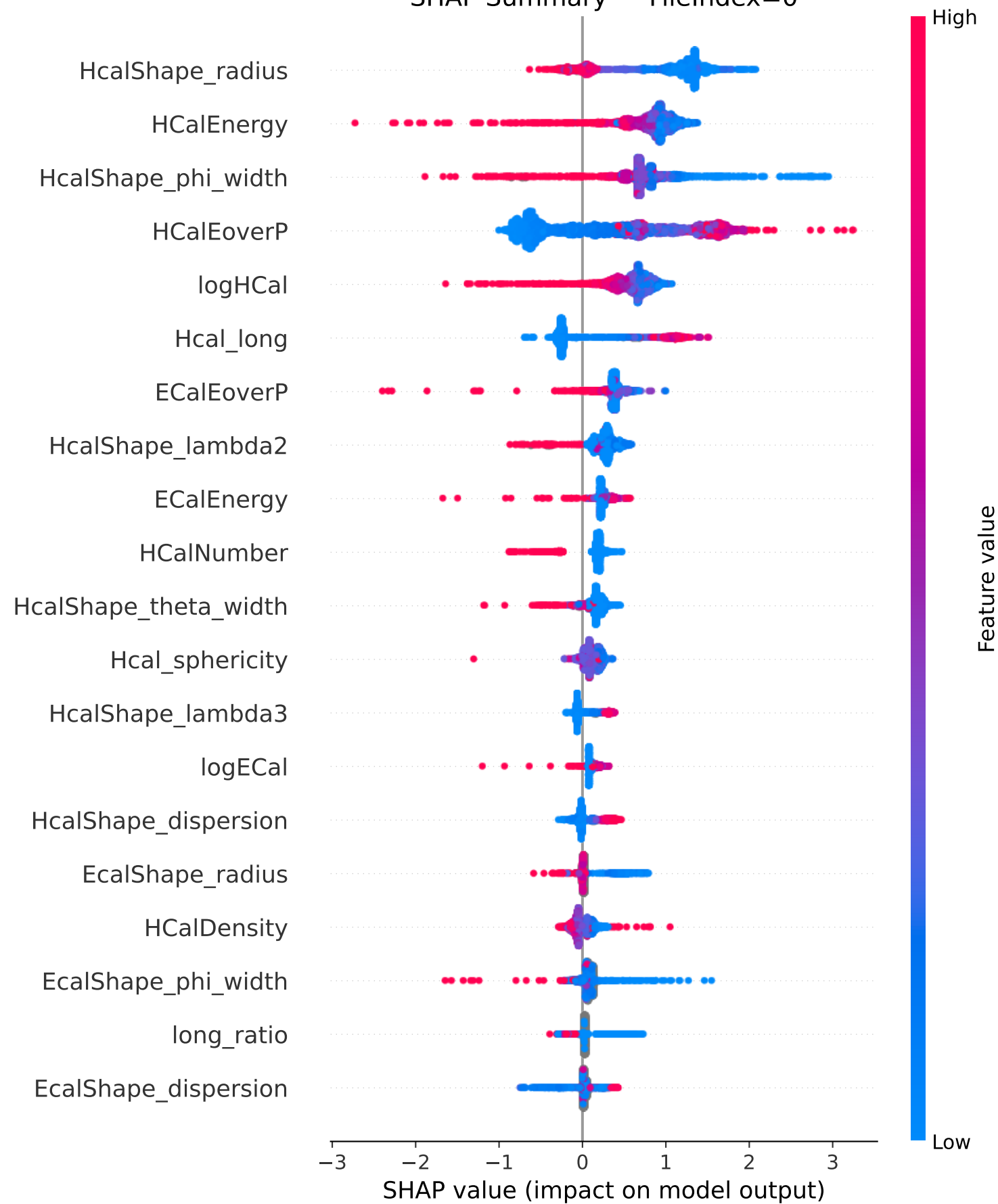
High

Feature value

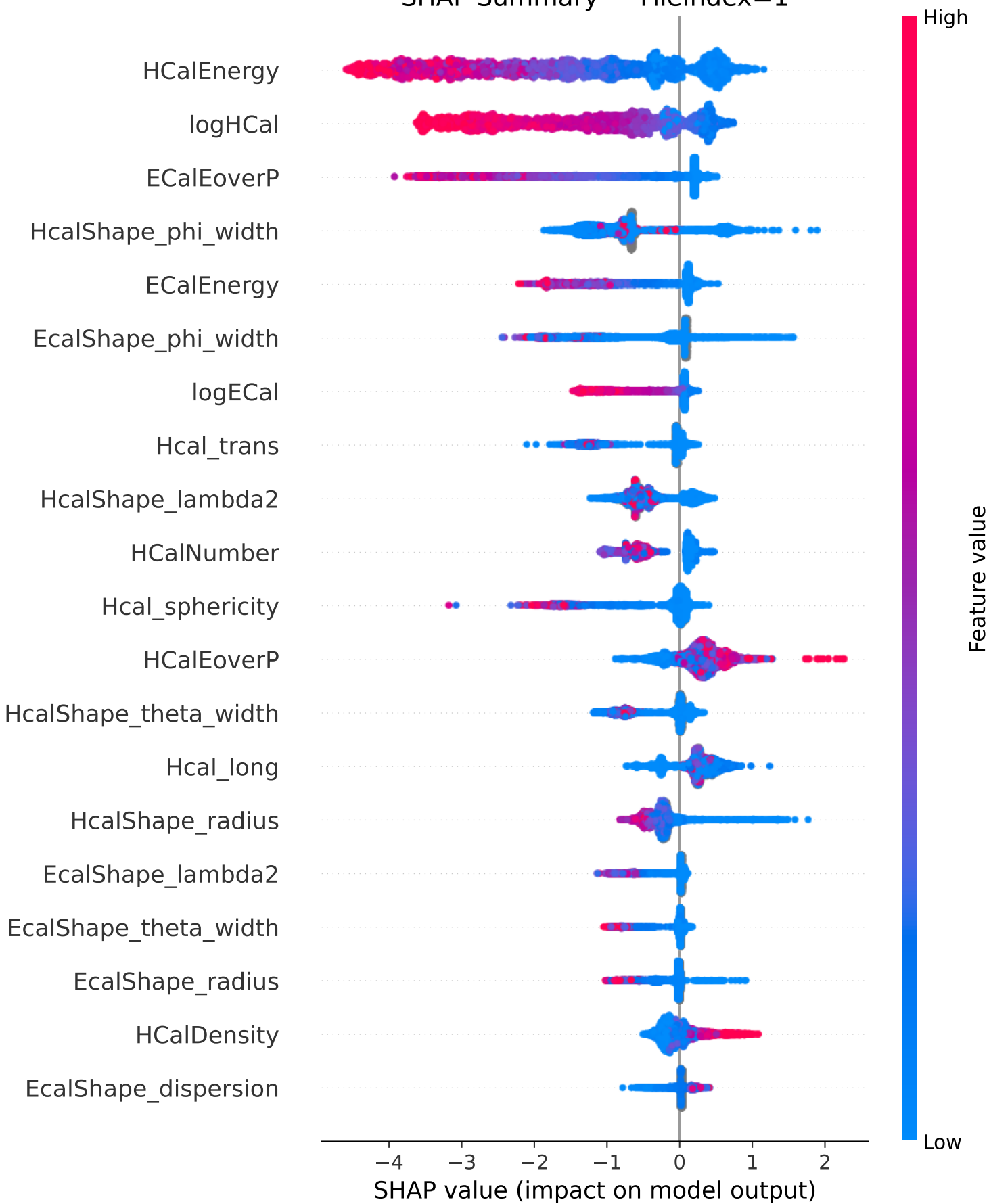
Low



SHAP Summary — FileIndex=0



SHAP Summary — FileIndex=1



SHAP Similarity Matrix — FileIndex vs FileIndex
(korelacja między wektorami ważności SHAP)

